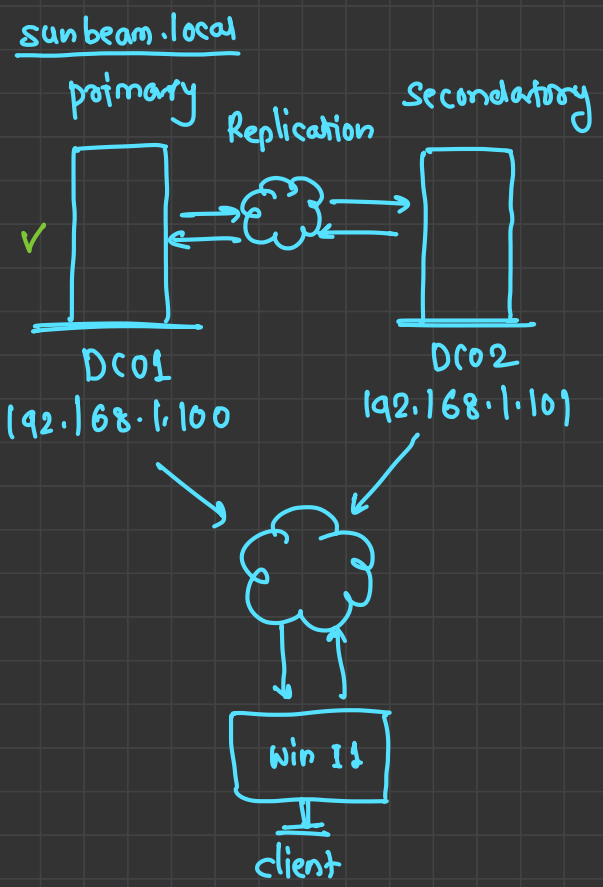
Additional Domain Controller


sunbeam.local
primary
Replication
Secondary
DC01
192.168.1.100
DC02
192.168.1.101
Win 11
client

Introduction

- An **Additional Domain Controller (ADC)** is a domain controller added to an **existing Active Directory domain** to provide redundancy, load balancing, and fault tolerance alongside the first (original) domain controller
- An ADC is any writable or read-only domain controller installed **after** the first DC has already created the domain
- It replicates the existing AD database from a current DC
- The first DC creates the domain; the ADC joins that existing domain as a helper and backup server
- The term is commonly used in practice to distinguish a newly added DC from the original (first) DC in a domain
- Technically, after installation, all writable DCs in a domain are peers and participate in multi-master replication
- **Why Add an ADC?**
  - **Fault tolerance** — if one DC fails, the domain continues to function
  - **Load balancing** — distributes authentication and directory requests across multiple servers
  - **Improved performance** — clients connect to a nearby DC, reducing network traffic
  - **Site support** — provide local authentication for branch offices or remote sites

Key Considerations

- **FSMO Roles:** Can all be placed on one reliable DC; move them during maintenance if needed
- **Global Catalog:** In a single-domain forest, make all DCs GCs — no planning needed
- **Replication Monitoring:** Use **Repadmin** to verify replication health after adding an ADC
- **Backup:** Even with multiple DCs, regularly back up **System State** using Windows Server Backup
- **Physical Separation:** Place DCs in different physical locations or power circuits to avoid single points of failure

Types of Domain Controllers

- **Writable DC:** Full read/write copy of AD database; participates in replication → default
- **Read-Only Domain Controller (RODC):** Read-only copy; designed for branch offices
- **Global Catalog (GC) Server:** Holds partial replicas of all forest domains
- **FSMO Role Holder:** Holds one or more single-master operation roles

Requirements for Adding an ADC

- **Administrative Requirements**
  - **Domain Admin** or **Enterprise Admin** credentials
  - If adding a new domain to a forest: **Enterprise Admin**
  - Local **Administrator** rights on the target server
- **Hardware Requirements**
  - Processor: 1.4 GHz 64-bit (2+ GHz multi-core)
  - RAM: 512 MB - 2 GB+ (4–8 GB for large domains)
  - Disk Space: 32 GB - 100 GB+ (depends on AD size)
  - Network: 100 Mbps- Gigabit Ethernet
- **Software and Configuration Requirements**
  - **Windows Server** (~~2008, 2008 R2, 2012, 2012 R2, 2016,~~ 2019, 2022, 2025)
  - **NTFS-formatted** volumes (do **not** use ReFS for AD database, logs, or SYSVOL)
  - **Static IP address** (strongly recommended)
  - **DNS configuration** pointing to an existing DC/DNS server
  - **Network connectivity** to existing DCs
  - **Time synchronization** within 5 minutes of existing DCs
  - **Sufficient disk space** for ntds.dit, logs, and SYSVOL
  - adprep run if required (forest/domain prep for new OS versions)

Required Ports for Replication

- **TCP/UDP 53**: DNS
- **TCP/UDP 88**: Kerberos authentication
- **TCP/UDP 389**: LDAP
- **TCP 135**: RPC Endpoint Mapper
- **TCP/UDP 445**: SMB (for SYSVOL replication)
- **TCP 3268/3269**: Global Catalog
- **TCP/UDP 464**: Kerberos password change
- **Dynamic RPC ports**: 49152-65535 (for RPC/DCOM)

Features of an Additional Domain Controller

- **Authentication:** Verifies user/computer credentials via Kerberos or NTLM
- **Authorization:** Controls access to network resources
- **AD Database Storage:** Hosts ntds.dit and SYSVOL
- **Replication:** Multi-master replication with all other DCs
- **Group Policy:** Distributes and enforces GPOs
- **DNS Integration:** AD-integrated DNS zones
- **Global Catalog:** Optional; forest-wide search capability
- **FSMO Roles:** Can hold one or more FSMO roles
- **LDAP Support:** Directory access via LDAP
- **Kerberos KDC:** Acts as Key Distribution Center
- **Site Awareness:** Clients find nearest DC
- **Load Balancing:** Distributes authentication requests

Backup and Recovery of ADC



- **System State Backup**
  - Registry
  - COM+ Class Registration Database
  - Boot files
  - Active Directory (on DCs)
  - SYSVOL
  - Certificate Services (on CAs)
  - Cluster database (on cluster nodes)
  - IIS metabase (on IIS servers)
- **Recovery types**
  - Non-Authoritative: Restores objects; allows replication to overwrite
  - Authoritative: Restores objects; marks them not to be overwritten
  - Bare Metal Recovery: Full system restore
  - AD Recycle Bin: Restores deleted objects (Server 2008 R2+)

Best Practices

- **Minimum two DCs per domain:** Fault tolerance
- **Make all DCs GCs in single-domain forests:** Simplifies design
- **Place DCs in different physical locations:** True fault tolerance
- **Use static IP addresses:** Reliable DNS registration
- **Regular System State backups:** Protect against logical corruption
- **Monitor replication daily:** Early problem detection
- **Keep time sync accurate:** Kerberos requires <5 min difference
- **Use RODCs for branch offices:** Security in low-physical-security sites
- **Document all FSMO role holders:** Quick recovery
- **Test restores regularly:** Verify backup integrity
- **Implement tiered administration:** Least privilege
- **Clean up metadata after seizure:** Allow old DC to rejoin

Install Over the Network



- Install the **Active Directory Domain Services** role
- Run the **AD DS Configuration Wizard**
- Choose **Add a domain controller to an existing domain**
- Select options: **DNS Server**, **Global Catalog**, and/or **RODC**
- Select a source DC to replicate from (or allow the wizard to choose)
- Set paths for database, logs, and SYSVOL
- Complete prerequisites check and install

Benefits

- Redundancy / Fault Tolerance
  - If one DC fails, others continue authenticating users
- Load Balancing
  - Authentication requests distributed across multiple DCs
- Better Performance
  - Local DCs reduce WAN traffic and logon times
- Site Awareness
  - Clients use nearest DC for faster authentication
- Disaster Recovery
  - Surviving DC preserves the domain if one server is lost
- Reduced Downtime
  - Maintenance can be performed on one DC while others serve clients

Disadvantages



- Increased Cost
  - Requires additional server hardware, licensing, and power
- Replication Traffic
  - More DCs = more replication traffic on the network
- Management Complexity
  - More servers to patch, monitor, and secure
- Diminishing Returns
  - Each additional DC adds only slight availability gains but higher costs
- USN Rollback Risk
  - Improper VM snapshots or restores can corrupt AD (use supported backups only)